

RESEARCH INTERESTS

My research interests lie in generalizable foundation models and improving pretrained networks through lightweight adaptation. I am broadly interested in representation learning, time series modeling, multimodality, reasoning, and information retrieval.

EDUCATION

Indian Institute of Technology, Bombay

MS by Research | Artificial Intelligence & Data Science

Research Assistant under Prof. Sunita Sarawagi on the project *Conversational AI over Spatio-Temporal Datasets*

July'22 - Present

CPI: 8.71*

University of Mumbai

BE | Computer Engineering

July'18 - May'22

CPI: 9.61

PUBLICATIONS

Afrin Dange, Vaibhav Raj, Praneeth Netrapalli, Sunita Sarawagi, "Transformer Designs for In-Context Learning in Foundation Models for Time Series Forecasting with Covariates" accepted at ICML 2024 Workshop on Theoretical Foundations of Foundation Models

RESEARCH EXPERIENCE

Lightweight Adaptation of Time Series Foundation Models

Aug'24 - Present

Thesis | Guide: Prof. Sunita Sarawagi

- Experimented with **Kernel-based methods** to improve zero-shot forecasting accuracy of SOTA time series **Foundation Models** including Moirai (Salesforce), TimesFM (Google), and Chronos (Amazon).
- Identified Moirai's limitations in utilizing exogenous variables and improved its forecasting by incorporating exogenous variables using our proposed two-stage **Lightweight Adapter** on VLDB Bench and EPF datasets. (submission under review)

Generalizing In-context Learning Across Sequence Length

Feb'24 - Present

Thesis | Guide: Prof. Sunita Sarawagi

- Investigated **in-context learning** capabilities across different **tokenization** and **attention** strategies in transformers.
- Evaluated the **length generalization** of these strategies on **linear regression**, **2-layer neural networks**, **decision trees**, and **Gaussian processes** by training transformers on sampled functions and evaluating on long sequences.
- Applied **distillation** along sequence length using methods such as first-k, top-k, most-recent-k, and random-k, and explored **stateful transformer** architecture to improve length generalization.

ICL-Enabled Foundation Model for Time Series

May'23 - June'24

Thesis | Guides: Prof. Sunita Sarawagi, Dr. Praneeth Netrapalli

- Explored a hybrid architecture for time series with **covariates** combining transformer layers with non-patched input to support **in-context learning** followed by patched input to better capture temporal patterns.
- Augmented existing time series with **synthetic covariates** generated via **Gaussian noising** for pretraining, achieving zero-shot performance comparable to a supervised patched transformer trained with covariates.
- Developed a custom pipeline for **distributed pretraining** with PyTorch **Fully Sharded Data Parallel** strategy.
- Conducted an empirical analysis on synthetic regression tasks revealing that **patching** hinders **in-context learning**.

Joint Modeling of Spatio-temporal Datasets

Jan'23 - April'23

MS Seminar & R&D | Guide: Prof. Sunita Sarawagi

- Conducted an exhaustive survey of existing techniques for **spatial representations** and **time series forecasting**.
- Automated downloading, **preprocessing**, and dataset management for large-scale spatiotemporal data from NDAP.
- Developed a multimodal foundation model combining time and spatial embeddings, and time series for **forecasting** and **missing value imputation**, focusing on leveraging related time series.

WORK EXPERIENCE

UPL Limited | Full Stack Developer Intern

June'21 - May'22

- Developed an e-commerce platform for Advanta (UPL seeds division) as part of a six-member team, handling backend integration with AWS Amplify and custom OTP authentication using Amazon Cognito and AWS SNS.

COURSE PROJECTS

Fact Extraction & VERification (FEVER) Task

Spring'24

CS 728: Organization of Web Information | Instructor: Prof. Soumen Chakrabarti

- Utilized **BM25**-based **sparse document-level retrieval** in the first level, and **Dense Passage Retrieval (DPR)** on the top-k documents in the second level for extracting evidence from **Wikipedia** pages.
- Fine-tuned a **GPT-2** model for **NLI** on claim-evidence pairs with **negative sampling**.
- Enhanced the two-stage retrieval by **indexing named entities** at the document level, leading to a **10.29%** increase in label accuracy and a **15.29%** improvement in evidence F1 score.

Cross-lingual In-context Learning in Multilingual LLMs

Spring'24

CS 728: Organization of Web Information | Instructor: Prof. Soumen Chakrabarti

- Conducted **cross-lingual in-context learning** experiments on XNLI and SMiLER **entity-relation extraction** tasks using encoder-decoder (**Flan-T5**) and decoder-only (**Llama-3**) models.
- Utilized semantically aligned few-shot examples leveraging **cosine similarity** measure on **mBERT** embeddings.
- Achieved a **19.5%** and **6.5%** average increase in **macro F1** scores for cross-lingual transfer from English to French and English to Russian on XNLI and SMiLER tasks, respectively, through semantically aligned few-shot examples.

Knowledge Graph Based Question Answering

Autumn'23

CS 635: Indexing, Retrieval and Learning for Text and Graphs | Instructor: Prof. Soumen Chakrabarti

- Developed a **multihop KGQA module** on the IMDb dataset by creating a **knowledge graph** using **Complex** for entity representation and fine-tuning **RoBERTa** to represent multihop relations based on the question.
- Achieved **Mean Reciprocal Rank (MRR)** with **MRR@1** of **0.25** and **MRR@10** of **0.35**.

Multilingual Hate Speech Detection

Autumn'22

CS 626: Speech, Natural Language Processing and the Web | Instructor: Prof. Pushpak Bhattacharyya

- Finetuned **mBERT** on the HateXplain dataset for English and Hindi, with 20% of training data translated to Hindi.
- Compared performance by fine-tuning with ground truth attention and evaluated **explainability** using **LIME**.

POSITION(S) OF RESPONSIBILITY

Teaching Assistant

Jan'25 - Present

CS 726: Advanced Machine Learning | Instructor: Prof. Sunita Sarawagi

Reviewer

Sept'24 - Present

- ICLR for submissions on Time Series Forecasting, In-Context Learning.

Department Coordinator

June'23 - June'24

Institute Student Companion Programme (ISCP) | IIT Bombay

- Coordinated the Institute PG Orientation for 1500+ entrants and Department Orientation, mentored juniors in academic and non-academic areas and completed the Mentor Training Programme by the Student Wellness Center.

SCHOLASTIC AND EXTRA-CURRICULAR ACHIEVEMENTS

2024 Delivered a talk on “**Conditional Generation Using GANs and Diffusion Models**” and “**Generative AI for Vision**” at an ISTE-approved Faculty Development Program hosted by VESIT and FCRIIT.

2022 Secured **99.33 percentile** amongst 77,257 candidates in **GATE CSE**

2020 Secured **1st Place** in **HackerEarth Hackathon: GAME JAM 1.0**

2020 Awarded **Best Female Volunteer** at NSS Camp

TECHNICAL SKILLS

Programming and Scripting Languages: Python, C++, SQL, Javascript, Bash

Libraries and Frameworks: PyTorch, Accelerate, Hugging Face Transformers, Scikit-Learn, cuML

Tools: AWS, Git, Docker, Weights & Biases, TensorBoard, L^AT_EX

KEY COURSES

- Introduction to Probability and Random Processes
- Speech, Natural Language Processing and the Web
- Indexing, Retrieval and Learning for Text and Graphs
- Deep Learning for NLP
- Advanced Machine Learning
- Foundations of Machine Learning
- Optimization in Machine Learning
- Organization of Web Information
- Introduction to AI, Data, and Policy

HOBBIES & INTERESTS

Learning Japanese on Duolingo (top 3% learner in 2024), Reading novels, Sketching